

Quiz 0
Algebra Review

Name: _____

MATH 2200, Fall 2019

1. Express as a single fraction and simplify the following:

(a) $\frac{\frac{1}{3} + \frac{2}{5}}{\frac{3}{2}}$

(b) $\frac{x}{y} \cdot \frac{s}{u} \cdot \frac{y}{s} \cdot \frac{u}{x}$

(c) $\frac{4}{x} - \frac{x+1}{xy} + \frac{x-2}{xz}$

(d) $\frac{7x}{3y} \cdot \frac{3y+2}{x}$

2. Use a counterexample to show that $\frac{a}{b+c} \neq \frac{a}{b} + \frac{a}{c}$.

3. Your friend informs you that two is equal to one and provides the following proof as justification:

$a = b$	
$a^2 = ab$	multiply both sides by a
$a^2 - b^2 = ab - b^2$	subtract b^2 from both sides
$(a - b)(a + b) = b(a - b)$	factor both sides
$a + b = b$	divide both sides by $a - b$
$b + b = b$	substitute a for b
$2b = b$	simplify the right hand side
$2 = 1$	divide both sides by b

Explain why this proof is invalid.